

Assignment 6 Report

Vigenère Cipher Cryptanalysis Using Kasiski Examination, Index of Coincidence and Frequency Analysis

Group Number: 15

Course/Lab: CryptoLab – Lab 6

Ciphertext: Ciphertext 1 – Odd Group Number

1. Aim

To implement and cryptanalyze a Vigenère cipher using Kasiski Examination, Index of Coincidence (IC), and frequency analysis to recover the key and plaintext.

2. Objectives

- Preprocess and clean the given ciphertext.
- Identify repeated patterns in the ciphertext.
- Estimate possible key lengths using Kasiski Examination.
- Use Index of Coincidence to determine the most probable key length.
- Divide the ciphertext into groups according to the estimated key length.
- Perform frequency analysis on each group.
- Determine the probable key using statistical analysis.
- Decrypt the ciphertext using the recovered key.
- Verify the recovered key by re-encrypting the plaintext.
- Confirm that the regenerated ciphertext matches the original ciphertext.

3. Problem Statement

The task is to cryptanalyze a Vigenère cipher without directly knowing its key. Kasiski Examination is used to identify repeated ciphertext patterns and their distances to obtain candidate key lengths. Index of Coincidence is then used to select the most probable key length. Frequency analysis is performed on the resulting ciphertext groups to recover the individual key letters and decrypt the original plaintext.

4. Ciphertext

The given ciphertext was cleaned by removing non-alphabetic characters and converting it to uppercase.

Total ciphertext length: 395 letters.

The first 60 characters of the cleaned ciphertext were:

DAZFISFSPAVQLSNPXYSZWXALCDAFGQUISMTPHZGAMKTTFTCCFXKFCRGGLPFE

The program successfully loaded 395 letters before beginning the cryptanalysis.

5. Methodology / Algorithm

The cryptanalysis was performed using the following steps:

1. Load the given ciphertext.
2. Clean the ciphertext and retain only alphabetic characters.
3. Find repeated sequences of characters in the ciphertext.
4. Calculate the distances between repeated patterns.
5. Factorize the distances to obtain possible key lengths.
6. Select the most frequent candidate key lengths from Kasiski Examination.
7. Calculate the average Index of Coincidence for different key lengths.
8. Select the key length whose IC is closest to the expected English IC.
9. Divide the ciphertext into groups according to the selected key length.
10. Perform frequency analysis on each group.
11. Calculate the most probable shift for every group.
12. Convert the shifts into key letters.
13. Combine the recovered letters to form the complete key.
14. Decrypt the ciphertext using the recovered key.
15. Re-encrypt the recovered plaintext using the same key.
16. Compare the regenerated ciphertext with the original ciphertext.
17. Report successful verification if both ciphertexts match.

6. Implementation

The implementation is divided into three Python modules:

File	Purpose
<code>vigenere_cipher.py</code>	Performs ciphertext cleaning, Vigenère encryption, decryption and verification.

<code>kasiski_analysis.py</code>	Performs repeated-pattern detection, distance calculation, factor analysis and Kasiski Examination.
<code>frequency_analysis.py</code>	Performs IC calculation, group splitting, frequency analysis and key recovery.
<code>main.py</code>	Executes the complete Vigenère cryptanalysis workflow.

7. Kasiski Examination

Kasiski Examination was performed on the 395-letter ciphertext.

The program found **25 repeated patterns**. Several repeated patterns produced useful distances such as 336, 330, 210, 294 and 112.

The collected distances were:

[14, 57, 98, 112, 114, 182, 210, 294, 330, 336]

The factor frequency analysis produced the following major candidate key lengths:

Key Length	Frequency
2	24
7	22
14	22
3	16
6	15
4	12
8	12
16	12
12	9
5	4
10	4

The top candidate key lengths obtained from Kasiski Examination were:

[2, 7, 14, 3, 6]

Thus, Kasiski Examination suggested **14** as one of the strongest candidate key lengths

8. Index of Coincidence Analysis

The Index of Coincidence was calculated for possible key lengths from 2 to 15.

The expected English IC was taken as approximately **0.0667**, while random text has an IC of approximately **0.0385**.

The important results were:

Key Length	Average IC	Difference from English IC
2	0.0448	0.0219
3	0.0430	0.0237
4	0.0429	0.0238
5	0.0416	0.0251
6	0.0441	0.0226
7	0.0508	0.0159
8	0.0435	0.0232
9	0.0410	0.0257
10	0.0443	0.0224
11	0.0427	0.0240
12	0.0412	0.0255
13	0.0403	0.0264
14	0.0644	0.0023
15	0.0407	0.0260

The highest and closest value to the expected English IC was obtained for **key length 14**.

Therefore:

Estimated Key Length = 14

The IC analysis also confirmed that 14 was one of the candidate lengths obtained from Kasiski Examination.

9. Frequency Analysis and Key Recovery

After determining the key length as 14, the ciphertext was divided into **14 groups**.

Frequency analysis was then performed independently on each group. For every group, the letter frequencies, percentages, IC value and most probable shift were calculated.

The recovered key letters were:

Group	Best Shift	Key Letter
1	0	A
2	12	M
3	1	B
4	17	R
5	14	O
6	8	I

7	18	S
8	4	E
9	19	T
10	7	H
11	14	O
12	12	M
13	0	A
14	18	S

Therefore, the recovered key is:

AMBROISETHOMAS

The program's frequency analysis output confirms the individual shifts and key letters for all 14 groups.

10. Recovered Key

The recovered Vigenère key is:

AMBROISETHOMAS

Key Length: 14

The key is repeated cyclically during encryption and decryption.

The first 28 characters of the key schedule were:

Key:

AMBROISETHOMASAMBROISETHOMAS

The corresponding plaintext and ciphertext values from the program output are:

Plain :

DOYOUKNOWTHELANDWHERE THE ORAN

Key :

AMBROISETHOMASAMBROISETHOMAS

Cipher:

DAZFISFSPA VQLSNPXY SZWXALCDAF

11. Decryption Result

Using the recovered key:

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the ciphertext was successfully decrypted.

The plaintext begins with:

DO YOU KNOW THE LAND WHERE THE ORANGE TREE BLOSSOMS

THE COUNTRY OF GOLDEN FRUITS AND MARVELOUS ROSES

The recovered plaintext continues as:

WHERE THE BREEZE IS OFTEN AND BIRDS LIGHTER

WHERE BEES GATHER POLLEN IN EVERY SEASON

AND WHERE SHINES AND SMILES LIKE A GIFT FROM GOD

AN ETERNAL SPRINGTIME UNDER AN EVER BLUE SKY

The complete decrypted plaintext contains **395 characters**. The program reports the recovered plaintext and key length explicitly.

12. Verification

After decryption, the recovered plaintext was re-encrypted using the recovered key:

AMBROISETHOMAS

The generated ciphertext was compared with the original ciphertext.

The result was:

Verification: MATCH -- Verification PASSED

Therefore, the recovered key and plaintext were successfully verified.

14. Screenshots

The experiment screenshots are stored in the repository under `ass6/screenshots/`.

They document the Kasiski Examination, Index of Coincidence, Decryption + recovered key Verification.

15. Conclusion

The Vigenère cipher was successfully cryptanalyzed using Kasiski Examination, Index of Coincidence and frequency analysis. Kasiski Examination produced several candidate key lengths, including 14. Index of Coincidence strongly supported key length 14 with an average IC of 0.0644, which was very close to the expected English IC of 0.0667.

After dividing the ciphertext into 14 groups and performing frequency analysis, the key was recovered as:

AMBROISETHOMAS

The ciphertext was successfully decrypted using the recovered key. Finally, re-encryption of the recovered plaintext produced the original ciphertext exactly, resulting in:

Verification: MATCH -- Verification PASSED

Thus, the Vigenère cryptanalysis implementation successfully recovered the key and plaintext and verified the result.